Arun Sekar

Data Science & Analytics | Generative AI | Supply Chain & Operations

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SUMMARY

Data Science & Analytics professional with 3+ years of experience delivering measurable business impact for Fortune 100 clients across retail and oil & gas. Experienced in building demand forecasting, inventory optimization, and Generative AI–powered decision systems. Strong in machine learning, statistical modeling, and multi-agent LLM workflows to drive operational efficiency and data-driven decisions.

PROFESSIONAL EXPERIENCE

Mu Sigma

Operations & Maintenance Analytics - Fortune 100 Oil & Gas — Sep 2022 – Present | Bengaluru

AI Optimized Work Planning

- Designed and implemented a **multi-agent LLM-powered Generative AI system** to automate maintenance work-order planning, enabling accurate prediction of required parts and reducing manual planner dependency.
- Built a Search Agent pipeline that dynamically generates queries from work-order planning fields and retrieves relevant historical work orders using Azure AI Search (semantic embeddings, keyword, and fuzzy search).
- Developed a Judge Agent using LLM-based relevance scoring to assess contextual and field-level similarity and curate high-quality few-shot examples for downstream decision-making.
- Engineered a Classifier and Predictor agent framework leveraging few-shot learning and LLM reasoning to classify work orders and generate precise part recommendations.
- Delivered measurable **business impact**, achieving **78% prediction accuracy**, reducing planner effort by **~75%**, and driving **~$5M annual savings** through faster planning and improved procurement decisions.

Supply Chain & Inventory Analytics - Nike

Inventory Health Evaluation

- Developed a 3-month **heuristic demand forecasting model** using bottom-up sales order aggregation and multi-source order book and sell-through data to improve short-term demand visibility and inventory planning accuracy.
- Built and optimized **regression-based ML models** (CatBoost, XGBoost) to predict cancellation rates, call-off probabilities, and sell-through, enabling dynamic forecast adjustments across regions.
- Designed and operationalized core inventory health metrics including Weeks of Supply (WOS) and Weeks on Hand (WOH) to evaluate inventory efficiency, coverage, and balance across Nike's global supply chain.
- Developed an inventory classification framework to segment products into Gainers and Drainers using WOS, WOH, and discount-based signals, improving supply–demand alignment and inventory disposition decisions.
- Implemented a **statistical backtesting and forecast bias-correction framework** using **t-score confidence bounds** and standard error analysis, improving forecast accuracy by **18%**, supported by Tableau dashboards.

SKILLS

Technical Skills

Python | SQL | Machine Learning | Predictive Modeling | Statistical Modeling | Time-Series Forecasting | Generative AI | RAG | Semantic Search | Multi-Agent Systems | Prompt Engineering | Data Analysis

Tools

AWS (S3, SageMaker) | Azure (Synapse Analytics, Azure OpenAI, Azure AI Search, DevOps, Data Lake) | Snowflake | Tableau | Visual Studio Code | Git | Excel | PowerPoint

ACHIEVEMENTS

Star Performer of the Team, *Award x 4*

Recognized for exceptional ownership, delivery excellence, rapid learning, and client trust across high-impact projects.

Hive Five Award, *Nike*
Recognized for exceptional collaboration and commitment, supporting cross-timezone delivery with WHQ teams and being acknowledged as a key asset to the project's success.

Education

B.Tech Chemical Engineering, 08/2018 – 05/2022
Kongu Engineering College, Anna University